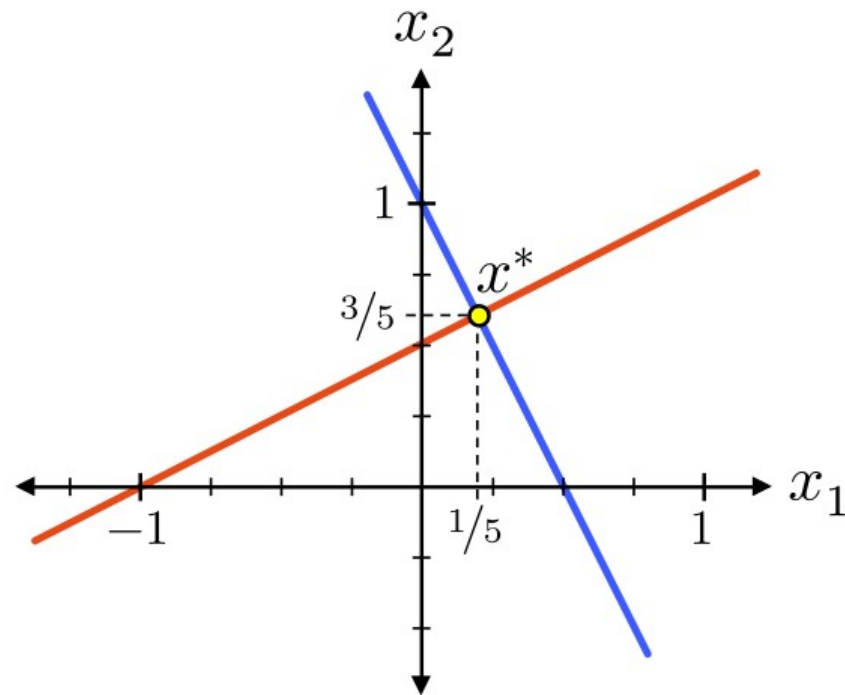


E7 Introduction to Computer Programming for Scientists and Engineers

Lecture week 13-B **Systems of linear equations**

System of linear equations

$$\begin{cases} 2x_1 + x_2 = 1 & \text{blue line} \\ -x_1 + 2x_2 = 1 & \text{red line} \end{cases}$$



Solution:

$$x^* = (x_1, x_2) = \left(\frac{1}{5}, \frac{3}{5} \right)$$

System of equations in function (a.k.a. matrix) form

The *matrix form* of a system of linear m equations is n unknowns:

$$Ax = b$$

The diagram illustrates the matrix form of a system of linear equations $Ax = b$. It shows the dimensions of the matrix A , the vector x , and the vector b .

- Matrix A is represented by a grid of dots. A vertical double-headed arrow to its left is labeled m , indicating the number of rows (equations). A horizontal double-headed arrow above it is labeled n , indicating the number of columns (unknowns).
- Vector x is shown as a column vector with elements $x_1, \vdots, x_n$. A vertical double-headed arrow to its right is labeled n , indicating its dimension.
- Vector b is shown as a column vector with elements $\cdot, \cdot, \cdot, \cdot, \cdot$. A vertical double-headed arrow to its right is labeled m , indicating its dimension.

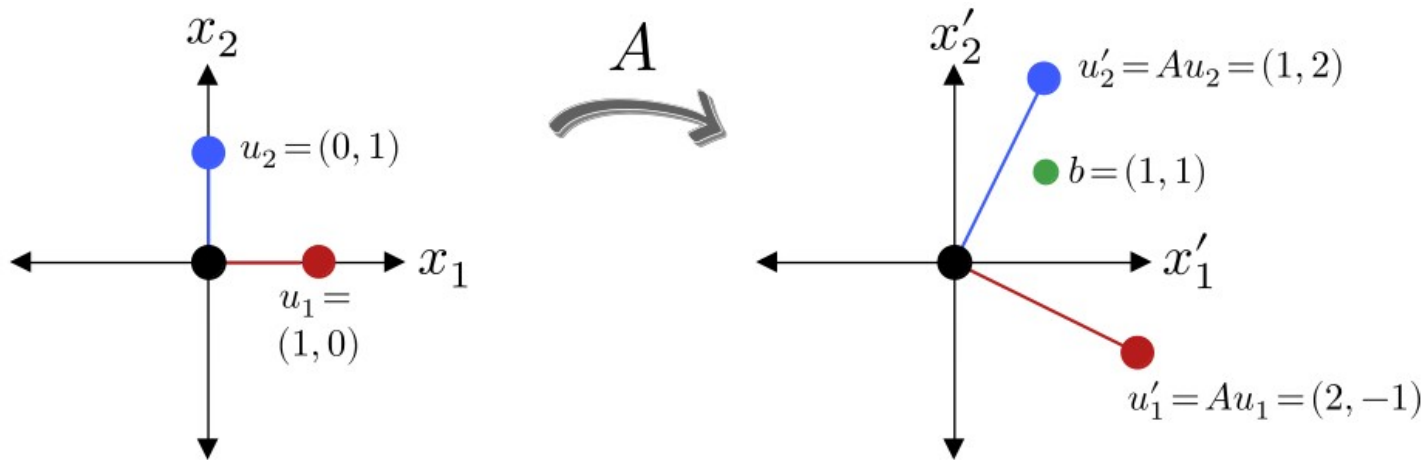
The equation is written as $Ax = b$, with the dimensions of each term clearly indicated by the arrows.

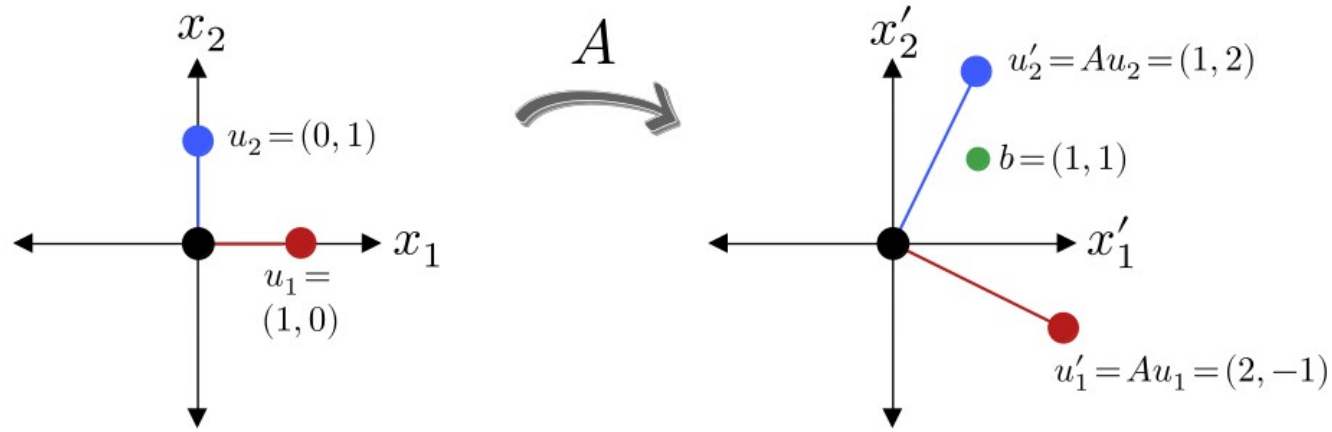
Example Put the following system of equations into matrix form.

$$\begin{cases} 2x_1 + x_2 = 1 \\ -x_1 + 2x_2 = 1 \end{cases}$$

$$\begin{bmatrix} 2 & 1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Function (or geometric) perspective:





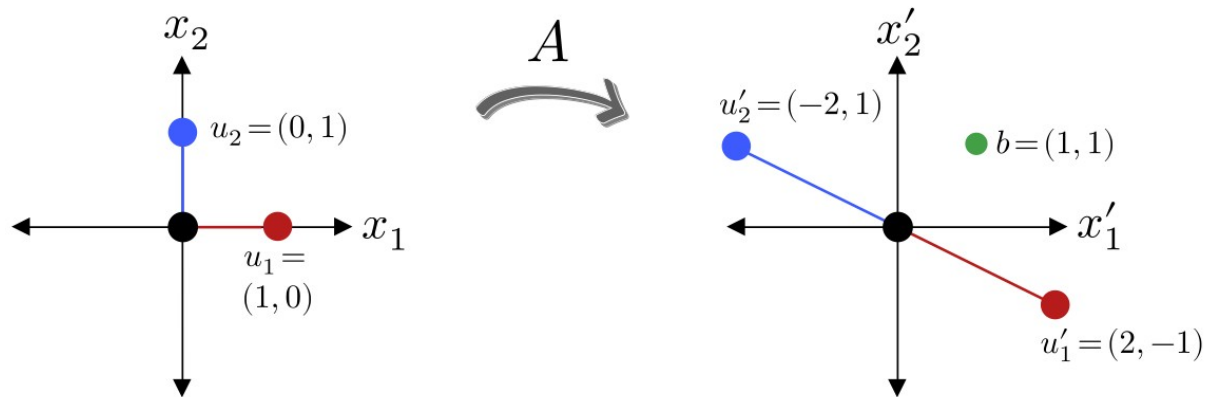
- *Span* of A :
 - The points of the codomain that are reachable via A .
 - The points that can be constructed as a combination of u'_1 and u'_2 .
- *Rank* of A : The dimension of the span of A .

NumPy : `rankA = np.matrix_rank(A)`

Example Two equations and two unknowns with no solution.

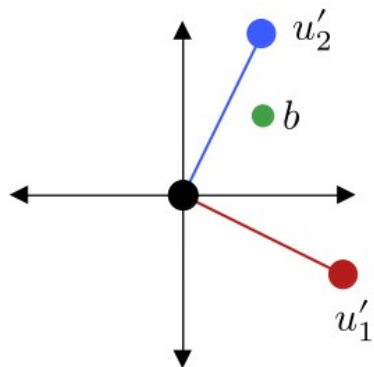
$$\begin{cases} 2x_1 - 2x_2 = 1 \\ -x_1 + x_2 = 1 \end{cases}$$

$$\begin{bmatrix} 2 & -2 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$



- u'_1 and u'_2 are colinear. Therefore rank of A equals 1.
- b is not in the span of A . Therefore there are no solutions.

Three cases

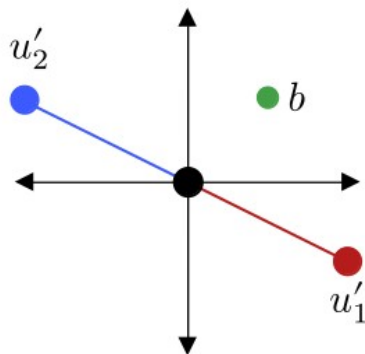


$$\text{rank}(A) = n$$



1 solution

well-determined

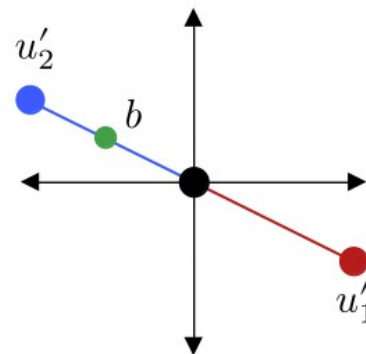


$$\begin{aligned} \text{rank}(A) &< n \\ b &\notin \text{span}(A) \end{aligned}$$



0 solutions

overdetermined



$$\begin{aligned} \text{rank}(A) &< n \\ b &\in \text{span}(A) \end{aligned}$$



∞ solutions

underdetermined

Augmented matrix $A|b$

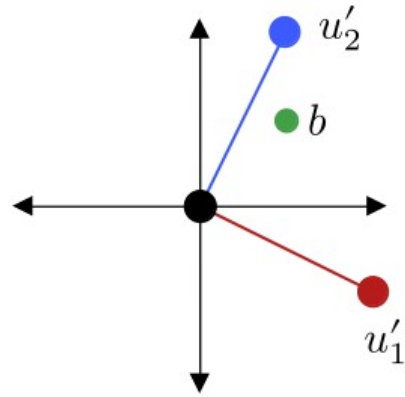
- The *augmented matrix* $A|b$ is obtained by horizontally stacking A and b .

$$A = \begin{bmatrix} 2 & -2 \\ -1 & 1 \end{bmatrix} \quad b = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$A|b = \begin{bmatrix} 2 & -2 & 1 \\ -1 & 1 & 1 \end{bmatrix}$$

NumPy : `Ab = np.hstack((A,b))`

The three cases in terms of the augmented matrix



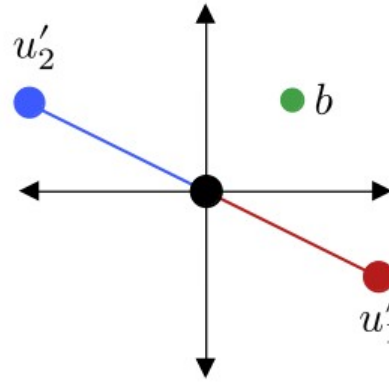
$$\text{rank}(A|b) = \text{rank}(A)$$

$$= n$$



1 solution

well-determined

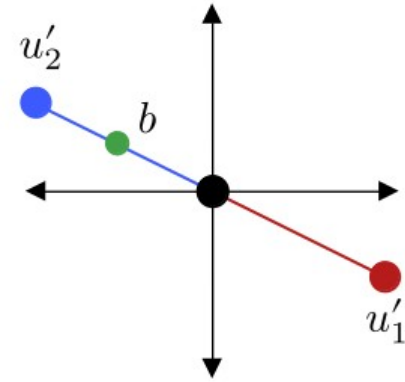


$$\text{rank}(A|b) > \text{rank}(A)$$



0 solutions

overdetermined



$$\text{rank}(A|b) = \text{rank}(A)$$

$$< n$$



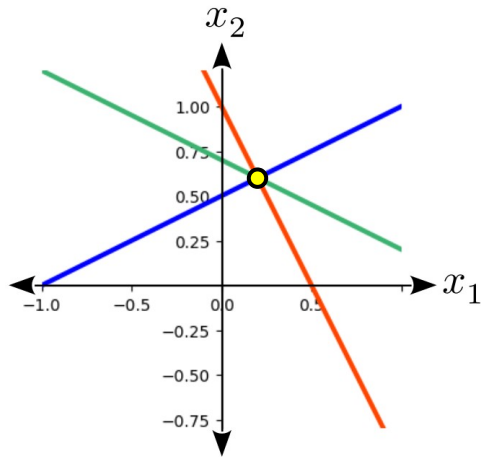
∞ solutions

underdetermined

Examples of equations, two unknowns $m = 3, n = 2$

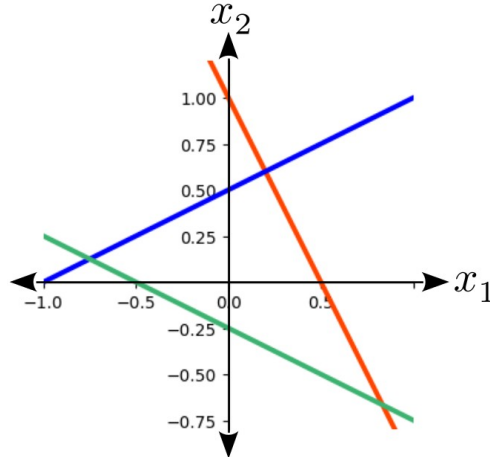
well-determined
(1 solution)

$$\begin{cases} 2x_1 + x_2 = 1 \\ -x_1 + 2x_2 = 1 \\ x_1 + 2x_2 = 1.4 \end{cases}$$



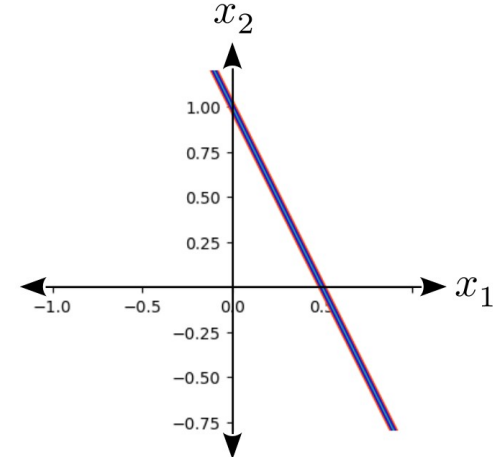
overdetermined
(0 solutions)

$$\begin{cases} 2x_1 + x_2 = 1 \\ -x_1 + 2x_2 = 1 \\ x_1 + 2x_2 = -0.5 \end{cases}$$



underdetermined
(∞ solutions)

$$\begin{cases} 2x_1 + x_2 = 1 \\ -2x_1 - x_2 = -1 \\ 4x_1 + 2x_2 = 2 \end{cases}$$

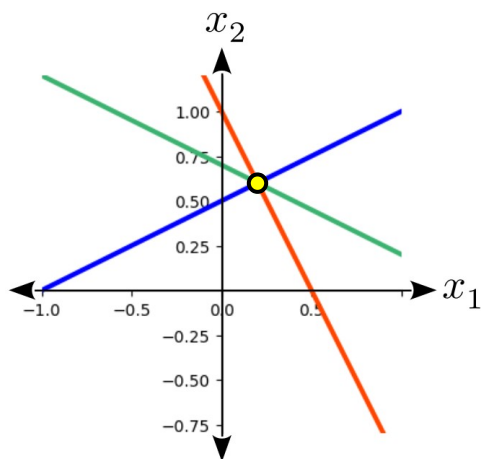


well-determined

$$A|b = \begin{bmatrix} 2 & 1 & 1 \\ -1 & 2 & 1 \\ 1 & 2 & 1.4 \end{bmatrix}$$

$$\text{rank}(A|b) = 2$$

$$\text{rank}(A) = 2$$

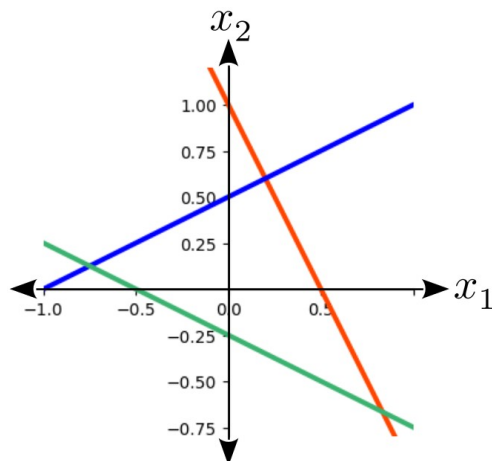


overdetermined

$$A|b = \begin{bmatrix} 2 & 1 & 1 \\ -1 & 2 & 1 \\ 1 & 2 & -0.5 \end{bmatrix}$$

$$\text{rank}(A|b) = 3$$

$$\text{rank}(A) = 2$$

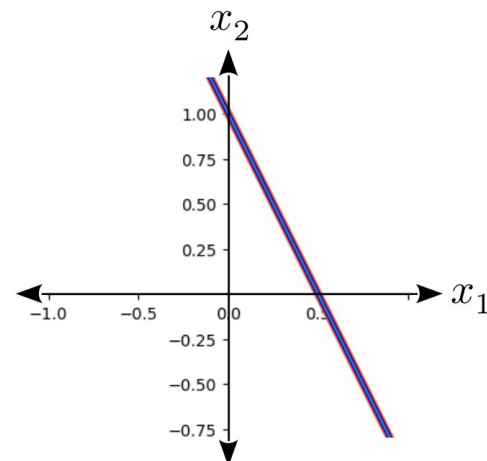


underdetermined

$$A|b = \begin{bmatrix} 2 & 1 & 1 \\ -2 & -1 & -1 \\ 4 & 2 & 2 \end{bmatrix}$$

$$\text{rank}(A|b) = 1$$

$$\text{rank}(A) = 1$$



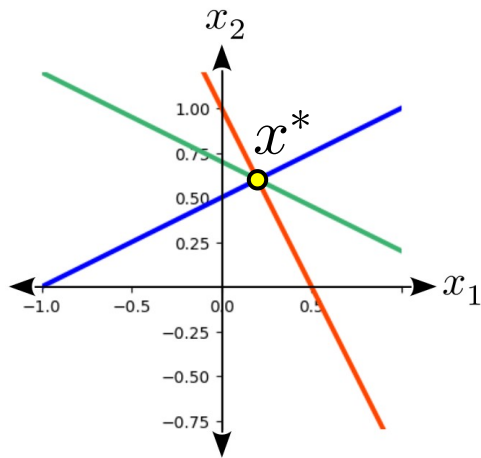
Moore-Penrose pseudoinverse A^+

$$x^* = A^+ b$$

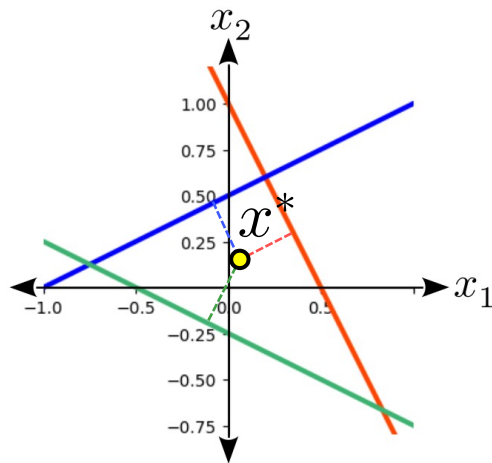
NumPy :

```
Ainv = np.pinv(A)
```

well-determined
(unique solution)



overdetermined
(least squares solution)



underdetermined
(minimum norm solution)

